

Linear Programming Report

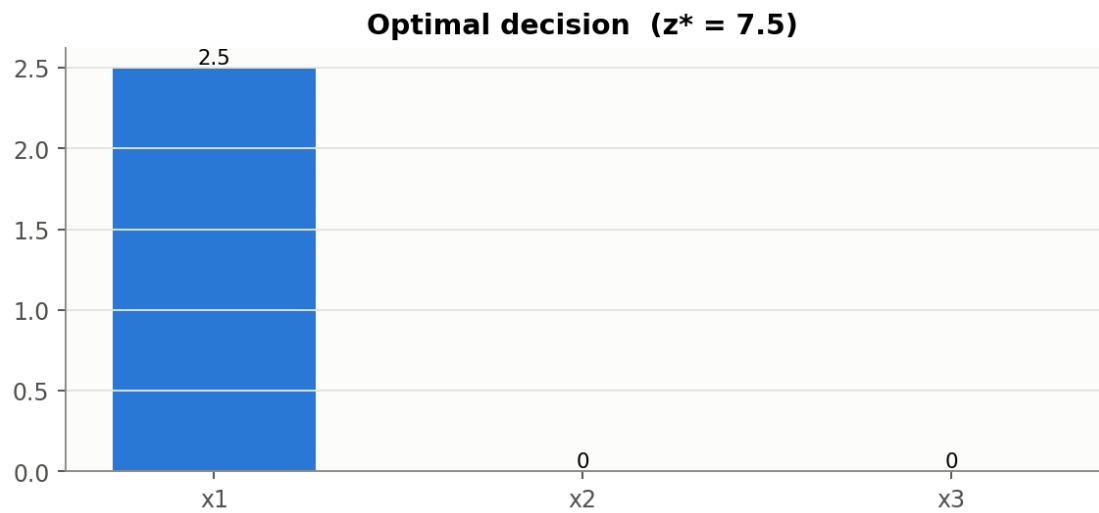
Sensitivity Analysis & Duality

Duality Example (min, >= form)

Canonical min problem with >= constraints and nonnegative variables. Its dual is a max problem with <= constraints and only two variables, so it can be solved graphically.

Source: Lecture slides 22-23

Problem type	MIN, 3 variables, 2 constraints
Status	OPTIMAL
Optimal Cost	7.5
Dual status	OPTIMAL
Prepared by	Dr. Islam Ali
Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
Date	27 September 2026, 15:11



1. Model formulation

MIN $z = 3x_1 + 2x_2 + 4x_3$
subject to
 $2x_1 + x_2 \geq 2$ (Con1)
 $2x_1 - x_2 + x_3 \geq 5$ (Con2)
 $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

Data table

Constraint	x1	x2	x3	Sense	RHS
Con1	2	1	0	>=	2
Con2	2	-1	1	>=	5
Objective c	3	2	4	min	—

Definitions

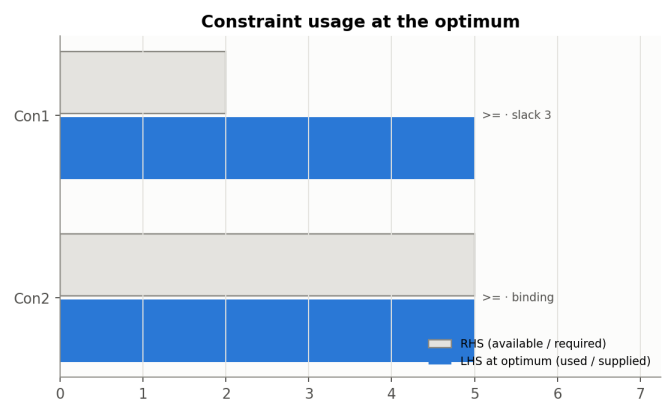
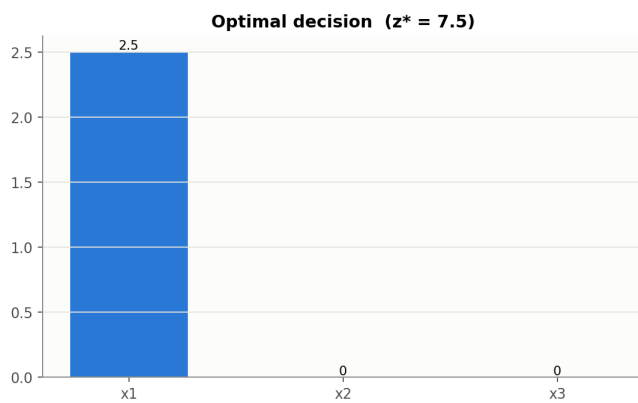
- **x1**: variable 1
- **x2**: variable 2
- **x3**: variable 3

2. Optimal solution

Optimal Cost: **7.5**

Variable	Description	Optimal value	Obj. coefficient	Contribution
x1	variable 1	2.5	3	7.5
x2	variable 2	0	2	0
x3	variable 3	0	4	0

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
Con1	5	>=	2	3	no
Con2	5	>=	5	0	yes



3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
x1	2.5	0	3	5	3	0	8
x2	0	3.5	2	∞	3.5	-1.5	∞
x3	0	2.5	4	∞	2.5	1.5	∞

Right-hand-side ranging (constraints)

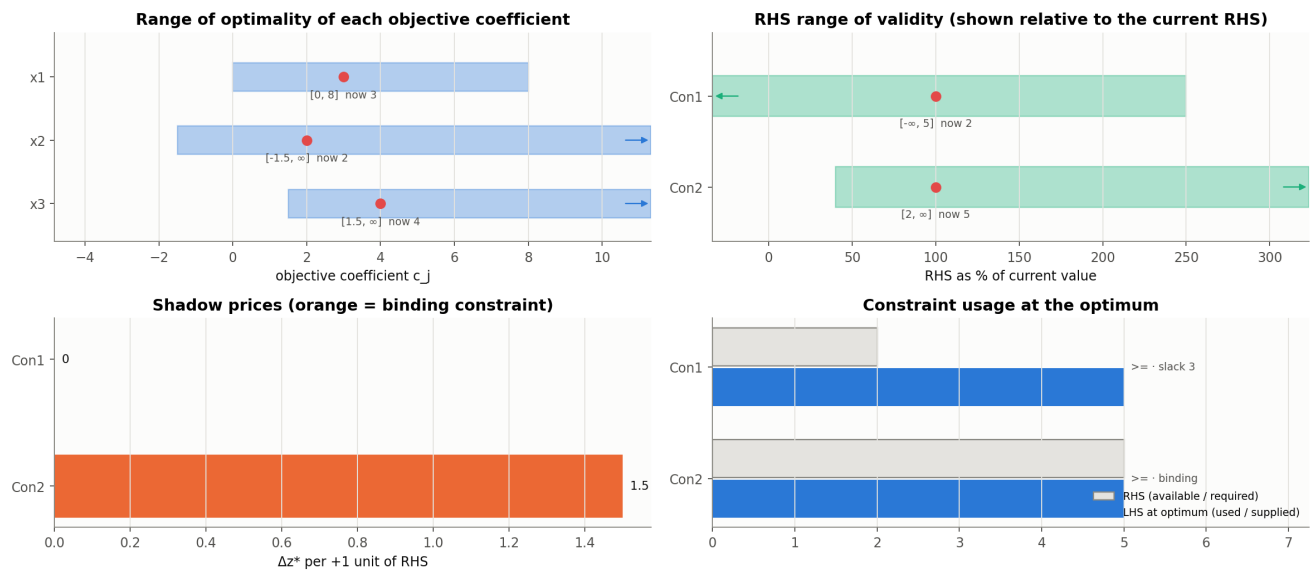
Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
Con1	5	$\geq$	2	3	0	3	∞	$-\infty$	5
Con2	5	$\geq$	5	0	1.5	∞	3	2	∞

Interpretation

- **x1** = 2.5. Its objective coefficient (3) may vary in $[0, 8]$ without changing the optimal solution (one change at a time).
- **x2** is not used (value 0). Reduced cost 3.5: its coefficient must decrease by about 3.5 before it becomes attractive.
- **x3** is not used (value 0). Reduced cost 2.5: its coefficient must decrease by about 2.5 before it becomes attractive.
- **Con1** is not binding (slack 3), so its shadow price is 0; the RHS can move within $[-\infty, 5]$ without effect on Cost.
- **Con2** is binding. Shadow price 1.5: each extra unit of RHS changes Cost by 1.5, valid while the RHS stays in $[2, \infty]$.

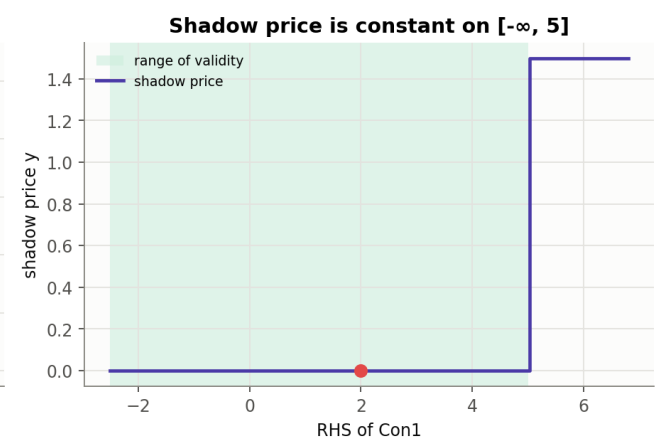
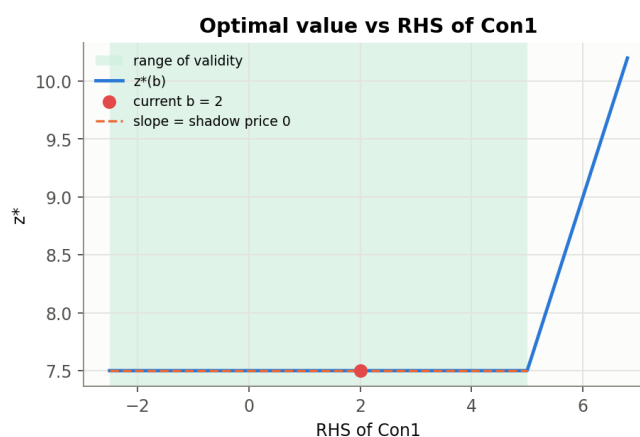
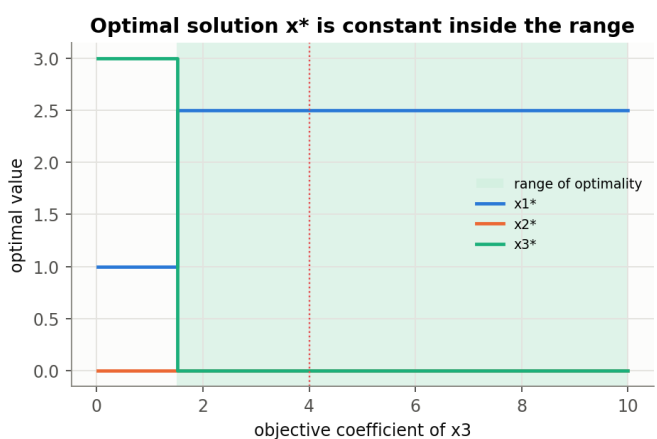
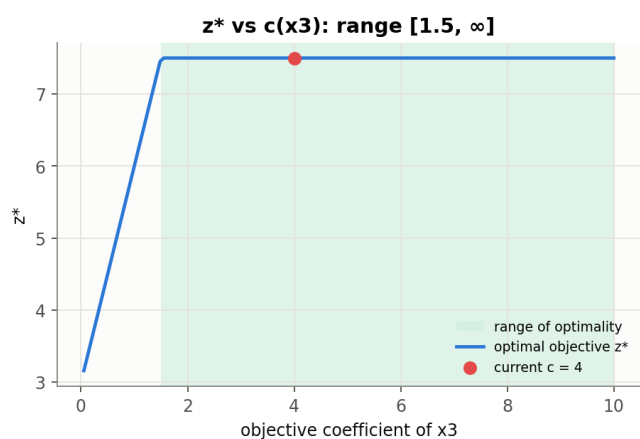
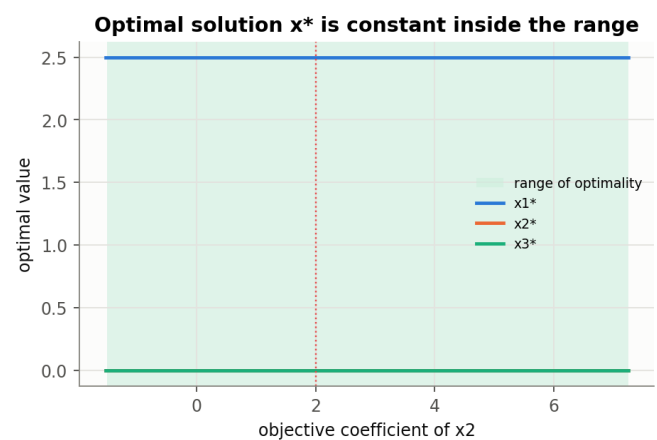
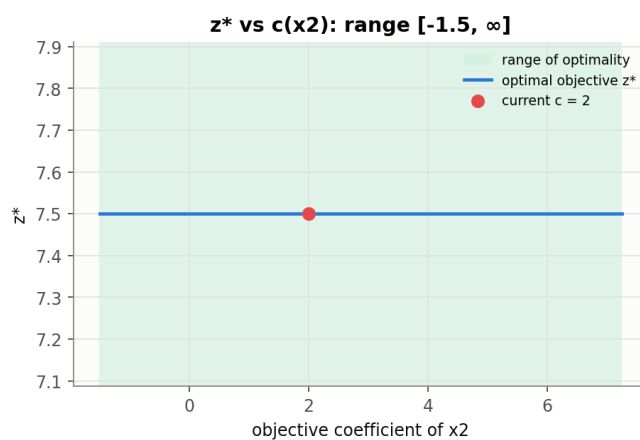
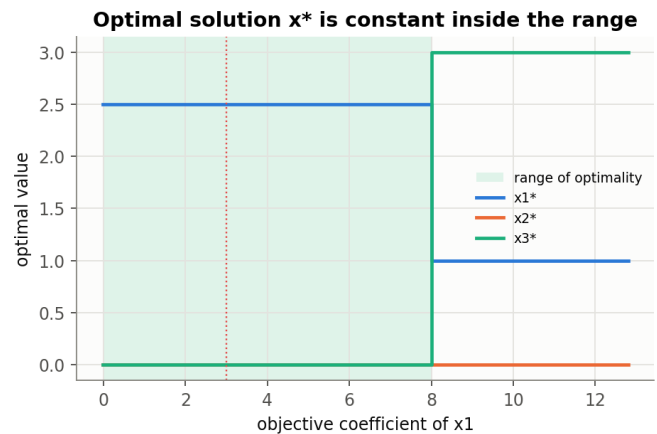
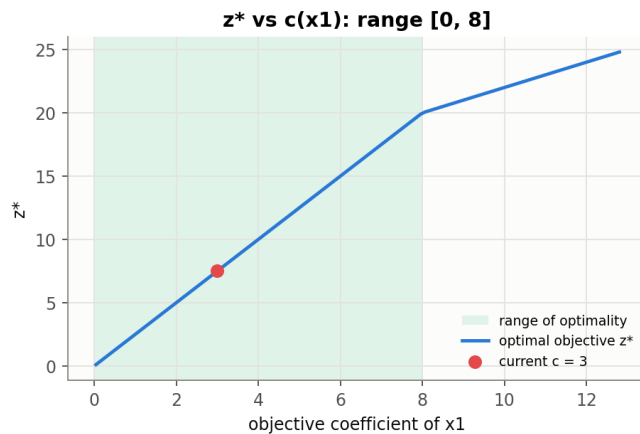
Sensitivity dashboard

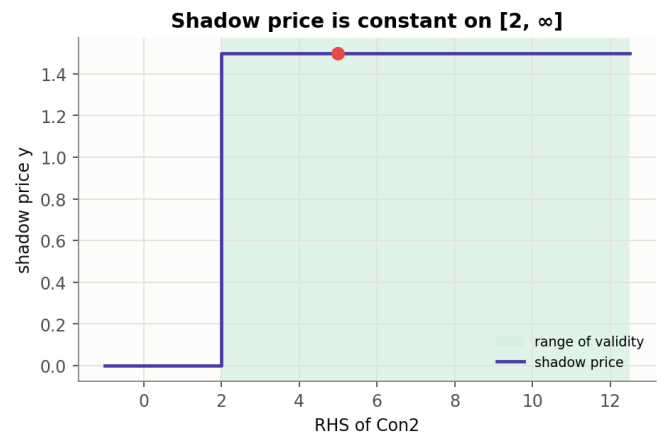
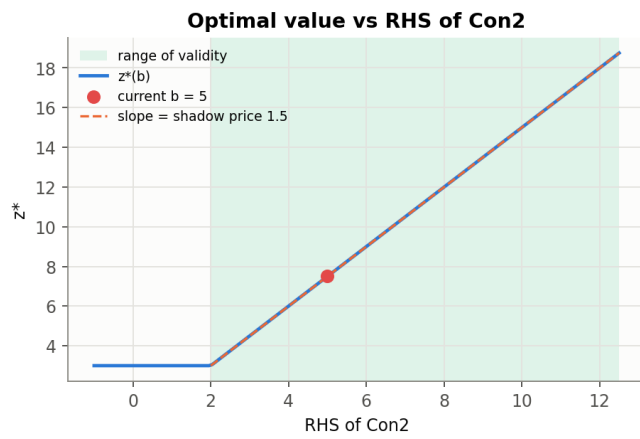
Duality Example (min, $\geq$ form): sensitivity dashboard ($z^* = 7.5$)



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.





6. Duality

Dual formulation

MAX $z = 2 \cdot y_{\text{Con1}} + 5 \cdot y_{\text{Con2}}$
 subject to
 $2 \cdot y_{\text{Con1}} + 2 \cdot y_{\text{Con2}} \leq 3$ (x1)
 $y_{\text{Con1}} - y_{\text{Con2}} \leq 2$ (x2)
 $y_{\text{Con2}} \leq 4$ (x3)
 $y_{\text{Con1}} \geq 0, y_{\text{Con2}} \geq 0$

Dual solution (resource / requirement prices)

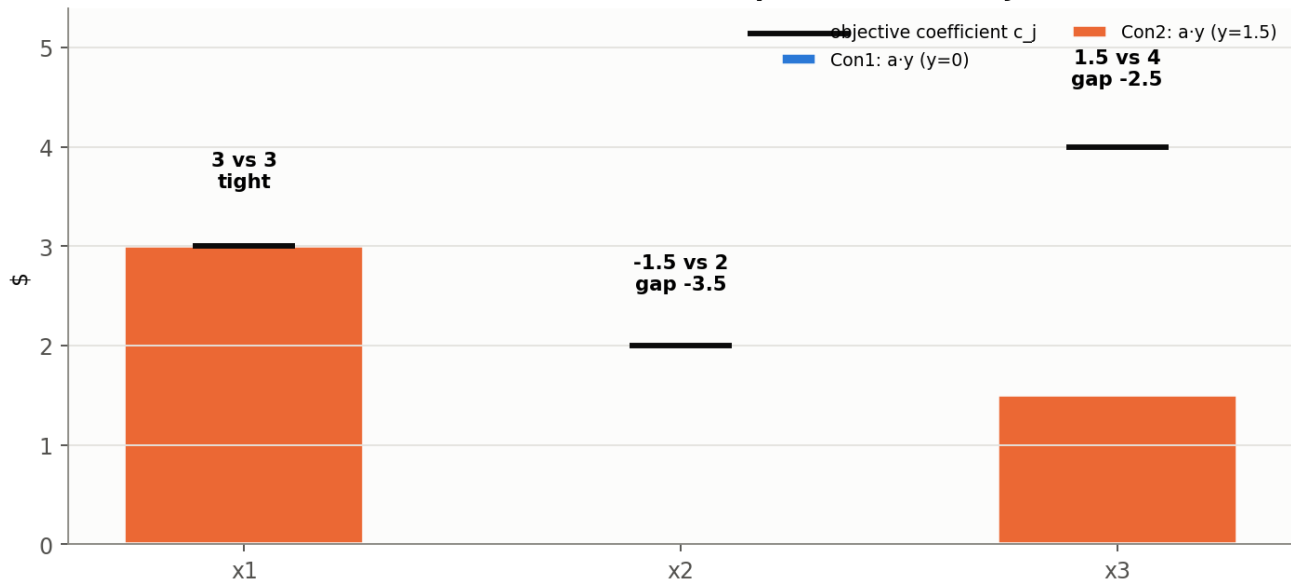
Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
y_{Con1}	Con1	0	0	0
y_{Con2}	Con2	1.5	1.5	7.5

Strong duality check: primal $z^* = 7.5$, dual $w^* = 7.5$ (equal ✓). Dual of the dual equals the primal: yes.

Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
Con1 slack $\times y_{\text{Con1}}$	3	0	0
Con2 slack $\times y_{\text{Con2}}$	0	1.5	0
$x1 \times$ surplus of dual constraint x1	0	2.5	0
$x2 \times$ surplus of dual constraint x2	3.5	0	0
$x3 \times$ surplus of dual constraint x3	2.5	0	0

Dual constraints: value of resources in each product vs its objective coefficient

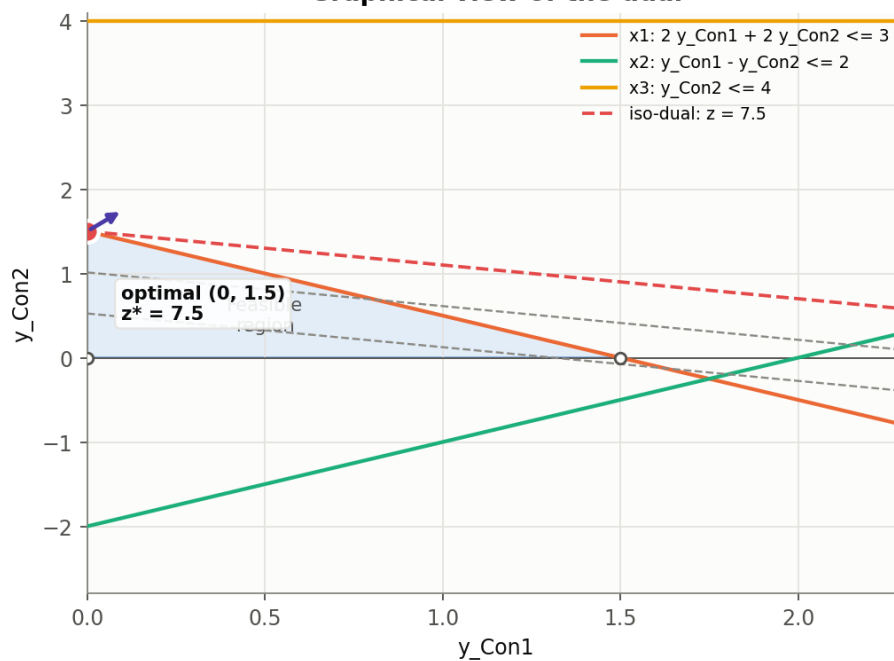


Primal (min): 2 constraints × 3 variables

3	2	4	← objective
2	1	0	
2	-1	1	
x_1	x_2	x_3	
			≥ 2
			≥ 5

Dual (max): 3 constraints × 2 variables

2	5	← objective
2	2	
1	-1	
0	1	
<i>y_Con1</i>	<i>y_Con2</i>	<= 3
		<= 2
		<= 4

Graphical view of the dual

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_j - c_j$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	1	x1	a1	7	yes	no	(0, 0, 0)
1	primal	1	x3	a2	3	yes	no	(1, 0, 0)
2	primal	1			-0	yes	yes	(1, 0, 3)
3	primal	2	e1	x3	15	yes	no	(1, 0, 3)
4	primal	2			7.5	yes	yes	(2.5, 0, 0)

Primal simplex - phase 1, tableau 0
enter x1, leave a1 (min ratio 1)

	x1	x2	x3	e1	e2	a1	a2	RHS
w	-4	0	-1	1	1	0	0	-7
a1	2	1	0	-1	0	1	0	2
a2	2	-1	1	0	-1	0	1	5

Primal simplex - phase 1, tableau 1
enter x3, leave a2 (min ratio 3)

	x1	x2	x3	e1	e2	a1	a2	RHS
w	0	2	-1	-1	1	2	0	-3
x1	1	0.5	0	-0.5	0	0.5	0	1
a2	0	-2	1	1	-1	-1	1	3

Primal simplex - phase 1, tableau 2
end of phase 1

	x1	x2	x3	e1	e2	a1	a2	RHS
w	0	0	0	0	0	1	1	0
x1	1	0.5	0	-0.5	0	0.5	0	1
x3	0	-2	1	1	-1	-1	1	3

Primal simplex - phase 2, tableau 3
enter e1, leave x3 (min ratio 3)

	x1	x2	x3	e1	e2	RHS
z	0	8.5	0	-2.5	4	-15
x1	1	0.5	0	-0.5	0	1
x3	0	-2	1	1	-1	3

Primal simplex - phase 2, tableau 4
optimal tableau

	x1	x2	x3	e1	e2	RHS
z	0	3.5	2.5	0	1.5	-7.5
x1	1	-0.5	0.5	0	-0.5	2.5
e1	0	-2	1	1	-1	3